PREDICTIVE MODELING AND PROFIT OPTIMIZATION FOR MULTI-CHANNEL RESTAU

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PREDICTIVE MODELING AND PROFIT OPTIMIZATION FOR MULTI-CHANNEL RESTAURANT OPERATIONS

Internship Project Report

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Submitted By

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ABSTRACT

The restaurant industry increasingly relies on multiple sales channels such as in-store dining, third-party delivery platforms, and self-delivery services. Managing these channels efficiently is essential for maximizing profitability while maintaining operational efficiency.

This project presents a Machine Learning-based solution for predicting restaurant profitability and optimizing business operations. Historical restaurant data was analyzed to identify key factors affecting profit, including order volume, average order value, commission rates, delivery costs, and channel distribution.

A predictive model was developed using Machine Learning techniques and integrated into an interactive Streamlit dashboard. The dashboard enables users to perform what-if analysis, visualize channel performance, evaluate business scenarios, and obtain actionable recommendations.

The developed system demonstrates how data-driven decision-making can support restaurant managers in improving profitability and operational performance.

Keywords: Machine Learning, Profit Optimization, Restaurant Analytics, Streamlit, Predictive Modeling, Business Intelligence.

ACKNOWLEDGEMENT

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I am thankful to my project mentor and team members for their valuable guidance, support, and encouragement throughout the project duration.

This project helped me gain practical knowledge in Machine Learning, Data Analytics, Business Intelligence, and Dashboard Development. The experience significantly enhanced my technical and problem-solving skills.

Finally, I would like to thank everyone who contributed directly or indirectly to the successful completion of this project.

Gauri Sanjivan Naikwadi.

CHAPTER 1: INTRODUCTION

1.1 Introduction

The restaurant industry has undergone significant transformation with the emergence of online food delivery platforms and digital ordering systems. Restaurants now operate through multiple sales channels such as in-store dining, Uber Eats, DoorDash, and self-delivery services. While these channels increase customer reach and revenue opportunities, they also introduce operational complexities and additional costs.

To remain competitive, restaurant operators must understand how factors such as order volume, average order value, delivery costs, commission rates, and channel distribution affect overall profitability. Traditional decision-making methods often rely on assumptions rather than analytical insights, leading to inefficient business strategies.

This project focuses on developing a predictive analytics system that uses Machine Learning techniques to estimate restaurant profitability and support operational decision-making. The system provides real-time profit predictions and enables managers to evaluate business scenarios before implementing changes.

1.2 Background

Data-driven decision-making has become essential in modern business environments. Machine Learning models can identify hidden relationships within operational data and generate accurate predictions. In the restaurant industry, predictive analytics can help optimize channel performance, reduce operational costs, and improve profit margins.

The integration of Machine Learning with interactive dashboards enables businesses to transform raw data into actionable insights, allowing managers to make informed decisions quickly and effectively.

CHAPTER 2: PROBLEM STATEMENT

Restaurants operating through multiple sales channels face several challenges in managing profitability. Each channel has different cost structures, commission fees, delivery expenses, and customer behaviors.

Without proper analytical tools, restaurant managers may struggle to understand:

- Which factors contribute most to profitability.
- How commission rates affect overall profit.
- The impact of delivery costs on operations.
- The optimal balance between delivery channels.
- Future profitability under changing business conditions.

The absence of predictive decision-support systems can result in inefficient resource allocation and reduced profitability.

Therefore, there is a need for a Machine Learning-based solution that can predict restaurant profit, analyze channel performance, and provide actionable recommendations to support strategic decision-making.

CHAPTER 3: OBJECTIVES

Primary Objective

To develop a Machine Learning-based Restaurant Profit Optimization System capable of predicting total net profit and supporting data-driven decision-making through an interactive dashboard.

Specific Objectives

1. To analyze restaurant operational data and identify factors affecting profitability.
2. To perform data preprocessing and feature engineering for predictive modeling.
3. To develop a Machine Learning model capable of predicting total restaurant profit.
4. To evaluate model performance using standard evaluation metrics such as MAE, RMSE, and R^2 Score.
5. To design and develop an interactive Streamlit dashboard for real-time profit prediction.
6. To implement What-If Analysis for evaluating business scenarios.
7. To visualize channel distribution and operational performance through interactive charts.
8. To provide business recommendations based on predictive insights.

Scope of the Project

The project focuses on restaurant profit prediction using historical operational data. It covers data preprocessing, exploratory data analysis, machine learning model development, dashboard implementation, and business scenario evaluation.

The solution is designed to assist restaurant managers in making informed decisions related to channel management, delivery operations, and profitability optimization.

CHAPTER 4: DATASET DESCRIPTION

4.1 Dataset Overview

The dataset used in this project is titled **"SkyCity Auckland Restaurants & Bars"**. The dataset contains restaurant operational and financial information collected across multiple sales channels.

The data includes order statistics, revenue information, delivery-related costs, commission rates, and profitability metrics.

Dataset Summary

Attribute	Description
Dataset Name	SkyCity Auckland Restaurants & Bars
Total Records	1696
Total Features	30
Data Type	Structured Tabular Data
Domain	Restaurant Operations & Profitability

4.2 Important Features

The following features were selected for predictive modeling:

Feature	Description
GrowthFactor	Business growth multiplier
AOV	Average Order Value
MonthlyOrders	Total monthly orders
InStoreOrders	Orders received in-store
UberEatsOrders	Orders received via Uber Eats
DoorDashOrders	Orders received via DoorDash
SelfDeliveryOrders	Orders delivered by restaurant
CommissionRate	Platform commission percentage

Feature	Description
DeliveryRadiusKM	Delivery service radius
DeliveryCostPerOrder	Cost incurred per delivery
InStoreShare	Percentage share of in-store orders
UE_share	Uber Eats share
DD_share	DoorDash share
SD_share	Self-delivery share

4.3 Target Variable

A new target variable named **TotalNetProfit** was created by combining profit generated from all delivery channels.

Formula:

TotalNetProfit=
InStoreNetProfit+
UberEatsNetProfit+
DoorDashNetProfit+
SelfDeliveryNetProfit

This variable serves as the prediction target for the Machine Learning model.

CHAPTER 5: DATA PREPROCESSING

5.1 Data Loading

The dataset was imported using the Pandas library and stored in a DataFrame for further analysis.

5.2 Data Inspection

Initial analysis included:

- Checking dataset dimensions
- Reviewing column names
- Verifying data types
- Identifying missing values

5.3 Feature Selection

Relevant operational and financial variables were selected based on their impact on restaurant profitability.

Selected features:

- GrowthFactor
- AOV
- MonthlyOrders
- InStoreOrders
- UberEatsOrders
- DoorDashOrders
- SelfDeliveryOrders
- CommissionRate
- DeliveryRadiusKM
- DeliveryCostPerOrder
- InStoreShare
- UE_share
- DD_share, SD_share

5.4 Feature Engineering

To improve predictive capabilities, a new variable called TotalNetProfit was generated by aggregating profit values from all channels.

Feature engineering helps transform raw business data into meaningful inputs for Machine Learning algorithms.

5.5 Data Preparation

The dataset was prepared for model training by:

- Selecting relevant features
- Defining the target variable
- Separating independent and dependent variables
- Splitting data into training and testing sets

Training Set: 80%

Testing Set: 20%

CHAPTER 6: EXPLORATORY DATA ANALYSIS (EDA)

6.1 Purpose of EDA

Exploratory Data Analysis was conducted to understand data distribution, identify patterns, and determine relationships between variables.

EDA helps improve feature selection and model performance.

6.2 Profit Distribution Analysis

A histogram was generated to visualize the distribution of TotalNetProfit across restaurants.

Observation

- Profit values were spread across a wide range.
- Most restaurants were concentrated within a moderate profit range.
- A few restaurants exhibited exceptionally high profitability.

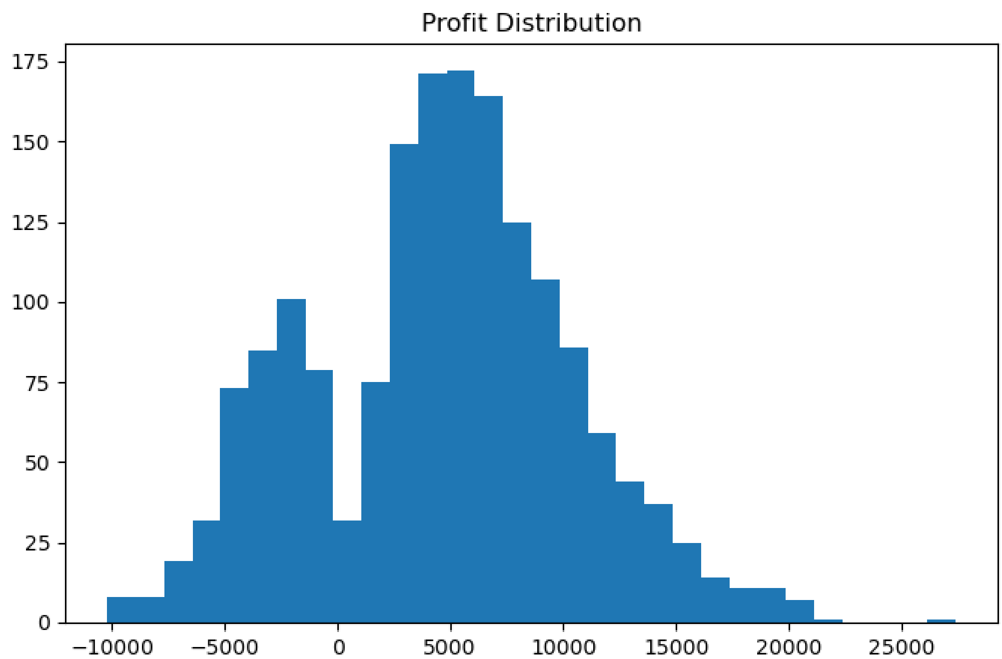

Figure 6.1: Distribution of Total Net Profit

6.3 Correlation Analysis

A correlation matrix was generated to measure relationships among variables.

Observation

- MonthlyOrders showed a positive correlation with profitability.
- Average Order Value (AOV) significantly influenced revenue generation.
- CommissionRate exhibited a negative impact on profit.
- Delivery-related features affected operational margins.

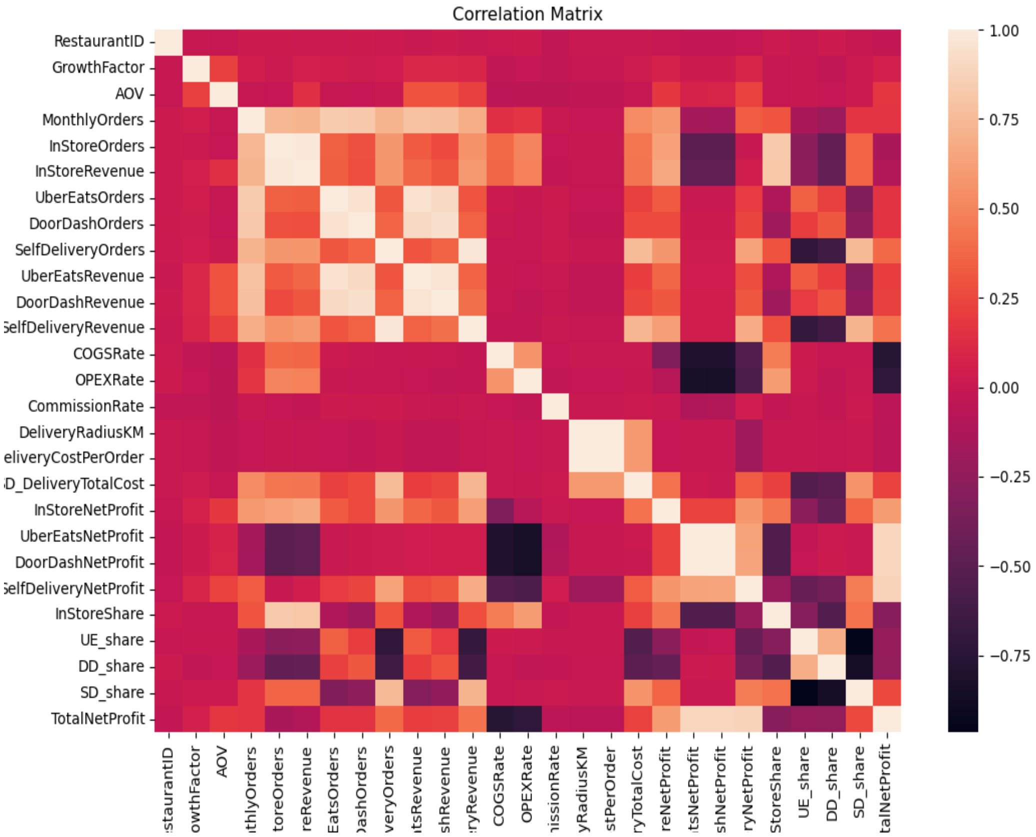

Figure 6.2: Correlation Matrix of Restaurant Variables

6.4 Feature Importance Analysis

Feature importance analysis was performed using the trained Machine Learning model.

The analysis identified the most influential factors affecting profitability.

Key Influencing Features:

- MonthlyOrders
- AOV
- GrowthFactor
- CommissionRate
- Channel Shares

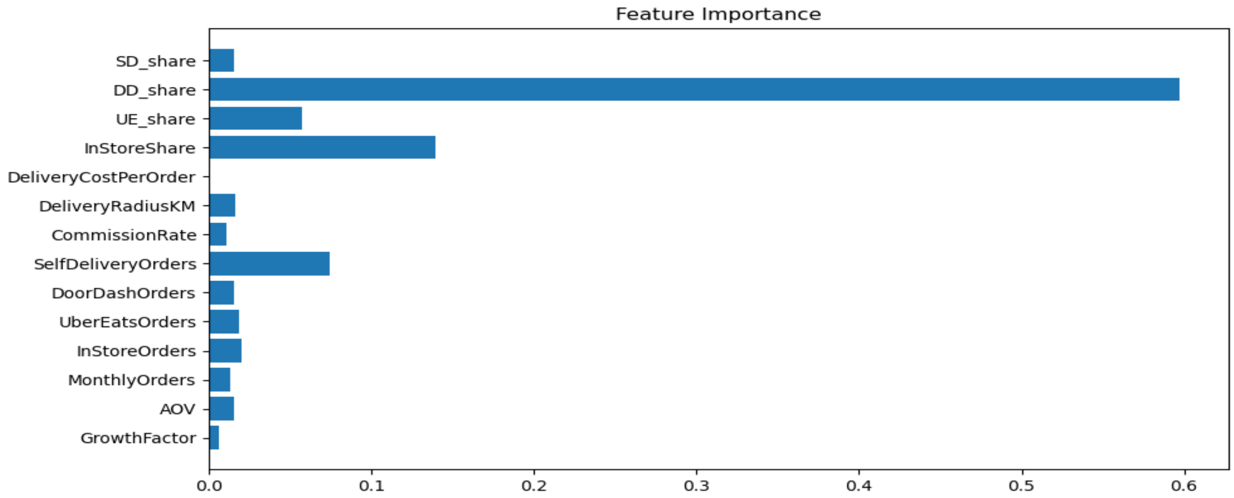

Figure 6.3: Feature Importance Analysis

6.5 EDA Findings

The exploratory analysis revealed that:

1. Order volume is a major driver of restaurant profitability.
2. Higher Average Order Values contribute positively to profit.
3. Commission charges significantly affect margins.
4. Channel distribution plays a critical role in operational performance.
5. Delivery costs influence overall profitability.

These insights were utilized during model development and dashboard design.

CHAPTER 7: MACHINE LEARNING MODEL DEVELOPMENT

7.1 Introduction

Machine Learning techniques were used to predict restaurant profitability based on operational and financial factors. The objective was to build a predictive model capable of estimating TotalNetProfit using historical restaurant data.

7.2 Feature Matrix and Target Variable

Input Features (X)

The following features were selected as model inputs:

- GrowthFactor
- AOV
- MonthlyOrders
- InStoreOrders
- UberEatsOrders
- DoorDashOrders
- SelfDeliveryOrders
- CommissionRate
- DeliveryRadiusKM
- DeliveryCostPerOrder
- InStoreShare
- UE_share
- DD_share
- SD_share

Target Variable (Y)

Target Variable:

TotalNetProfit

This variable represents the combined profit generated through all restaurant channels.

7.3 Train-Test Split

The dataset was divided into:

Dataset	Percentage
Training Data	80%
Testing Data	20%

The training dataset was used for model learning, while the testing dataset was used for performance evaluation.

7.4 Model Selection

The XGBoost Regressor algorithm was selected due to its ability to handle complex relationships within structured business datasets.

Advantages of XGBoost

- High predictive accuracy
- Efficient handling of large datasets
- Ability to capture non-linear relationships
- Feature importance extraction
- Reduced overfitting

7.5 Model Training

Training process:

1. Load dataset
2. Select features
3. Define target variable
4. Split data into training and testing sets
5. Train XGBoost Regressor
6. Evaluate model performance
7. Save trained model

The trained model was stored as:

profit_model.pkl

for integration with the Streamlit dashboard.

CHAPTER 8: MODEL EVALUATION

8.1 Introduction

Model evaluation was performed to measure prediction accuracy and overall performance.

The following evaluation metrics were used:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

8.2 Mean Absolute Error (MAE)

MAE measures the average difference between predicted and actual profit values.

Formula:

$MAE = (1/n)\ \Sigma\ |Actual - Predicted|$

Lower MAE values indicate better model performance.

8.3 Root Mean Squared Error (RMSE)

RMSE penalizes larger prediction errors and provides insight into model accuracy.

Formula:

$RMSE = \sqrt{[(1/n)\ \Sigma\ (Actual - Predicted)^2]}$

Lower RMSE values indicate improved prediction capability.

8.4 R^2 Score

The R^2 Score indicates how well the model explains variance in the target variable.

Formula:

$R^2 = 1 - (\text{Residual Sum of Squares} / \text{Total Sum of Squares})$

Interpretation:

- $R^2 = 1 \rightarrow$ Perfect prediction
- $R^2 = 0 \rightarrow$ No predictive capability

8.5 Evaluation Results

Insert your actual model results here:

Metric	Value
MAE	2714.26
RMSE	3803.71
R^2 Score	0.5502

Observation

The trained model achieved satisfactory predictive performance and successfully captured relationships between operational variables and profitability.

The results indicate that Machine Learning can be effectively applied to restaurant profit prediction and optimization.

CHAPTER 9: DASHBOARD DEVELOPMENT

9.1 Introduction

An interactive dashboard was developed using Streamlit to provide real-time access to profit predictions and business insights.

The dashboard allows restaurant managers to modify operational parameters and instantly observe their impact on profitability.

9.2 Dashboard Architecture

Workflow:

Restaurant Inputs
↓
Machine Learning Model
↓
Profit Prediction
↓
Interactive Visualization
↓
Business Recommendations

9.3 Dashboard Features

1. Profit Prediction

The dashboard predicts Total Net Profit using user-defined operational parameters.

2. KPI Cards

Key Performance Indicators displayed:

- Predicted Profit
- Commission Rate
- Monthly Orders

These metrics provide a quick overview of business performance.

3. Input Summary

A summary table displays all selected business parameters used for prediction.

4. Channel Distribution Visualization

An interactive donut chart displays the percentage contribution of:

- InStore Orders
- Uber Eats
- DoorDash
- Self Delivery

This visualization helps managers understand channel dependency.

5. What-If Analysis

The dashboard evaluates hypothetical business scenarios.

Example:

If Commission Rate increases by 5%, the dashboard automatically calculates the expected change in profitability.

This feature supports strategic decision-making.

6. Business Recommendation Engine

The dashboard provides recommendations based on:

- Channel mix
- Delivery strategy
- Platform dependency

These recommendations help improve operational efficiency and profitability.

9.4 Dashboard Screenshots

Insert the following screenshots:

Figure 9.1 Dashboard Homepage

Figure 9.2 KPI Cards

Figure 9.3 Channel Distribution Chart

Figure 9.4 What-If Analysis

Figure 9.5 Business Recommendation Section

9.5 Benefits of the Dashboard

- Supports real-time decision-making
- Improves business visibility
- Evaluates profitability scenarios
- Simplifies operational analysis
- Enhances strategic planning

CHAPTER 10: RESULTS AND FINDINGS

10.1 Project Results

The developed Machine Learning model successfully predicted restaurant profitability using operational and financial parameters. The integration of the model with an interactive Streamlit dashboard enabled real-time profit estimation and business scenario analysis.

The system provided a user-friendly interface for evaluating operational decisions and understanding their impact on profitability.

10.2 Key Findings

The analysis revealed several important business insights:

1. Order Volume Drives Profitability

Monthly order volume was identified as one of the most significant factors influencing restaurant profit. Restaurants with higher order volumes generally achieved better profitability.

2. Average Order Value (AOV) Impacts Revenue

Higher Average Order Values contributed positively to revenue generation and overall profit.

3. Commission Rates Affect Margins

Third-party delivery platforms impose commission charges that directly reduce restaurant profitability.

4. Channel Distribution Influences Performance

The balance between InStore, Uber Eats, DoorDash, and Self Delivery channels plays an important role in determining overall business performance.

5. Data-Driven Decisions Improve Outcomes

The What-If Analysis feature demonstrated how predictive analytics can assist managers in evaluating operational strategies before implementation.

10.3 Dashboard Outcomes

The dashboard successfully provided:

- Real-time profit prediction
- Interactive channel analysis
- Operational scenario evaluation
- Business recommendations
- Data export functionality

The solution transformed raw operational data into actionable business intelligence.

CHAPTER 11: CONCLUSION

The objective of this project was to develop a predictive analytics system capable of estimating restaurant profitability and supporting operational decision-making.

The project successfully utilized Machine Learning techniques to analyze restaurant data and predict Total Net Profit based on key operational variables. Exploratory Data Analysis helped identify significant business factors, while feature engineering improved the predictive capabilities of the model.

The Streamlit dashboard provided an intuitive platform for real-time prediction, channel analysis, and What-If Analysis. The recommendation engine further enhanced decision-making by offering actionable business insights.

Overall, the project demonstrated the effectiveness of Machine Learning and Business Intelligence tools in solving real-world restaurant management challenges.

The developed solution can assist restaurant operators in improving profitability, optimizing channel strategies, and making informed business decisions.

CHAPTER 12: FUTURE SCOPE

Although the current system provides valuable business insights, several enhancements can be implemented in future versions.

Future Enhancements

1. Real-Time Data Integration

The dashboard can be connected to live restaurant databases and APIs for real-time profit monitoring.

2. Advanced Forecasting

Time-series forecasting models can be incorporated to predict future revenue and profitability trends.

3. Customer Behavior Analysis

Customer segmentation and purchasing behavior analysis can be integrated to improve marketing strategies.

4. Inventory Optimization

Inventory management features can be added to minimize waste and improve operational efficiency.

5. Multi-Location Support

The system can be extended to support restaurant chains operating across multiple locations.

6. Cloud Deployment

The dashboard can be deployed on cloud platforms such as AWS, Azure, or Streamlit Community Cloud for remote accessibility.

7. AI-Powered Recommendations

Advanced Artificial Intelligence techniques can be used to generate more personalized and intelligent business recommendations.

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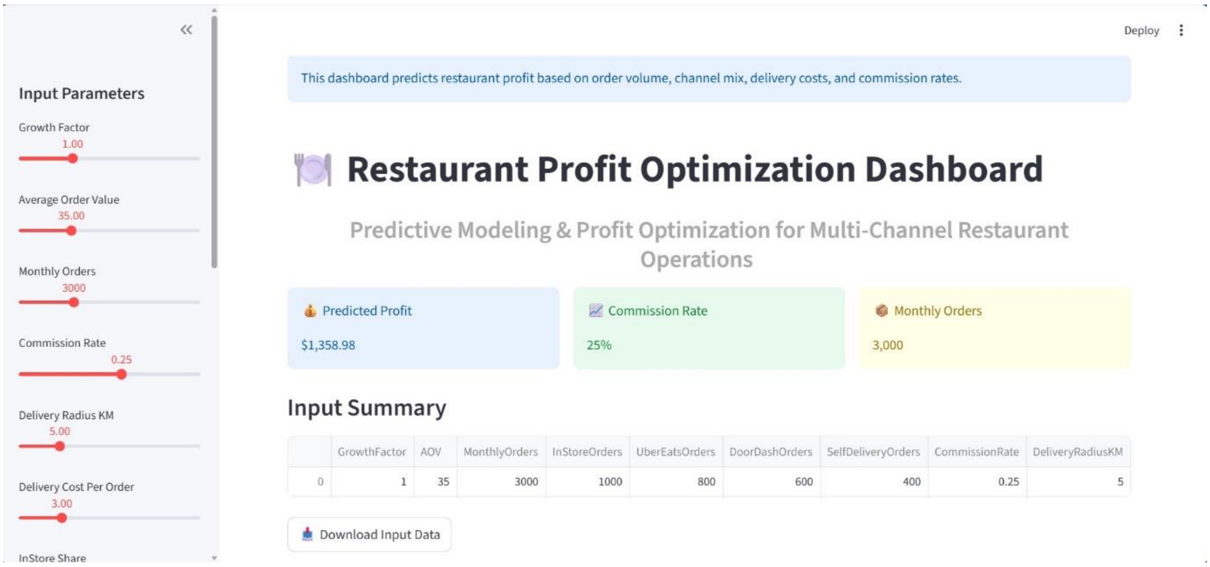
APPENDIX

Appendix A: Dashboard Screenshots

Figure A.1: Restaurant Profit Optimization Dashboard Homepage



Description:
This figure shows the main dashboard interface developed using Streamlit. The dashboard allows users to input operational parameters and obtain real-time profit predictions.

Figure A.2: Key Performance Indicators (KPI Cards)

Description:
The KPI section displays important business metrics including Predicted Profit, Commission Rate, and Monthly Orders. These metrics provide a quick overview of restaurant performance.

Figure A.3: Input Summary Table


Input Summary

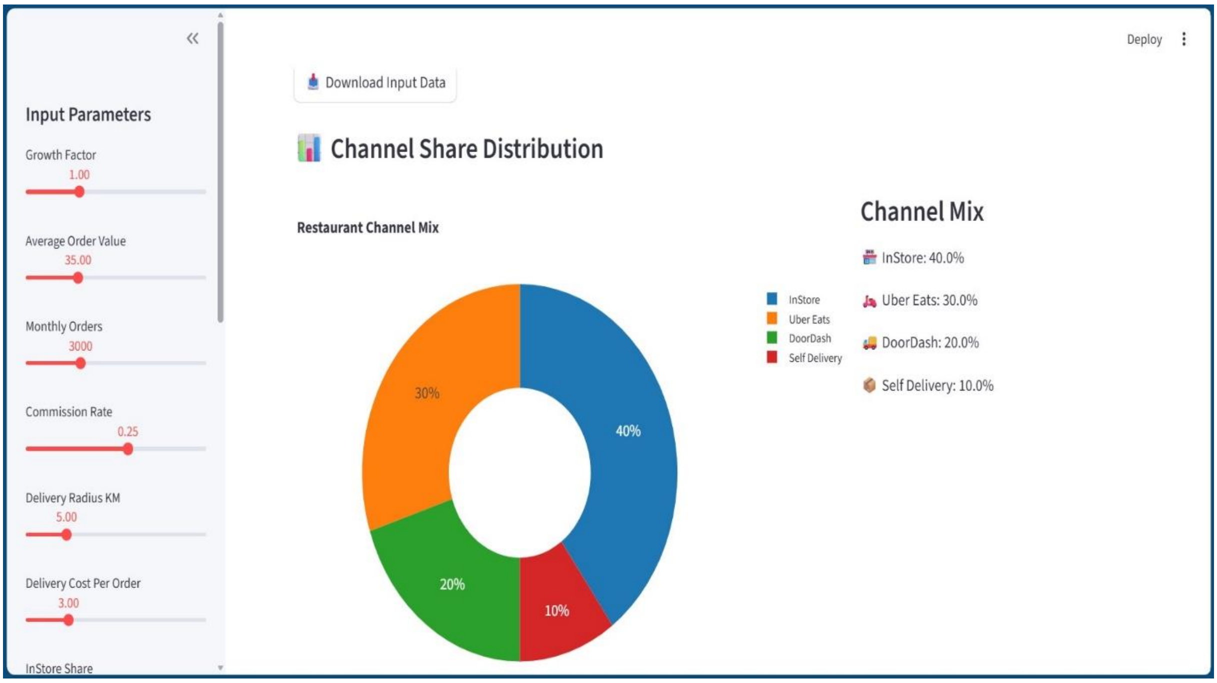
	GrowthFactor	AOV	MonthlyOrders	InStoreOrders	UberEatsOrders	DoorDashOrders	SelfDeliveryOrders	CommissionRate	DeliveryRadiusKM
0	1	35	3000	1000	800	600	400	0.25	5

Download Input Data

Description:
This table displays all user-selected operational parameters used as input for the Machine Learning model.

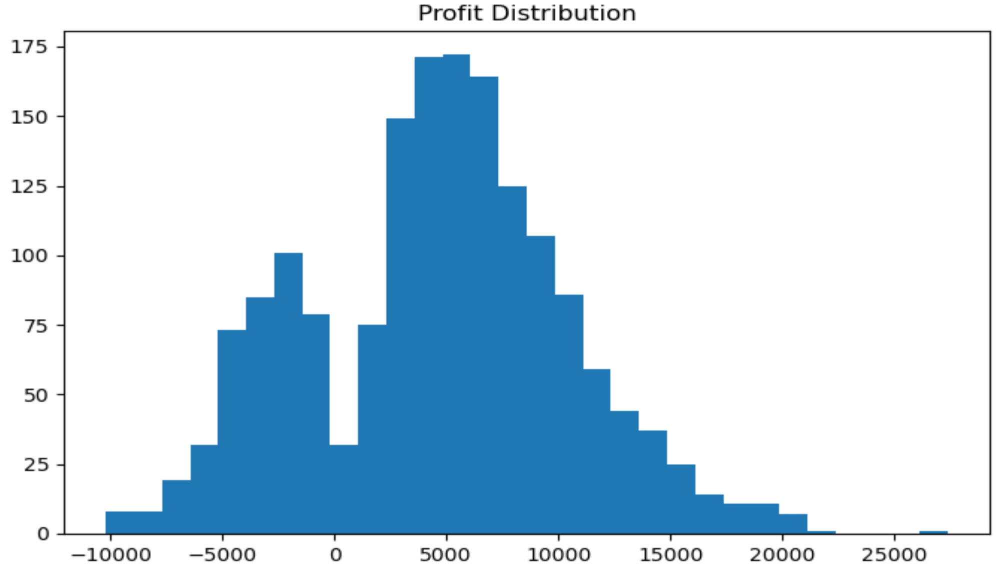
Appendix B: Visualization Outputs1

Figure B.1: Channel Share Distribution Chart



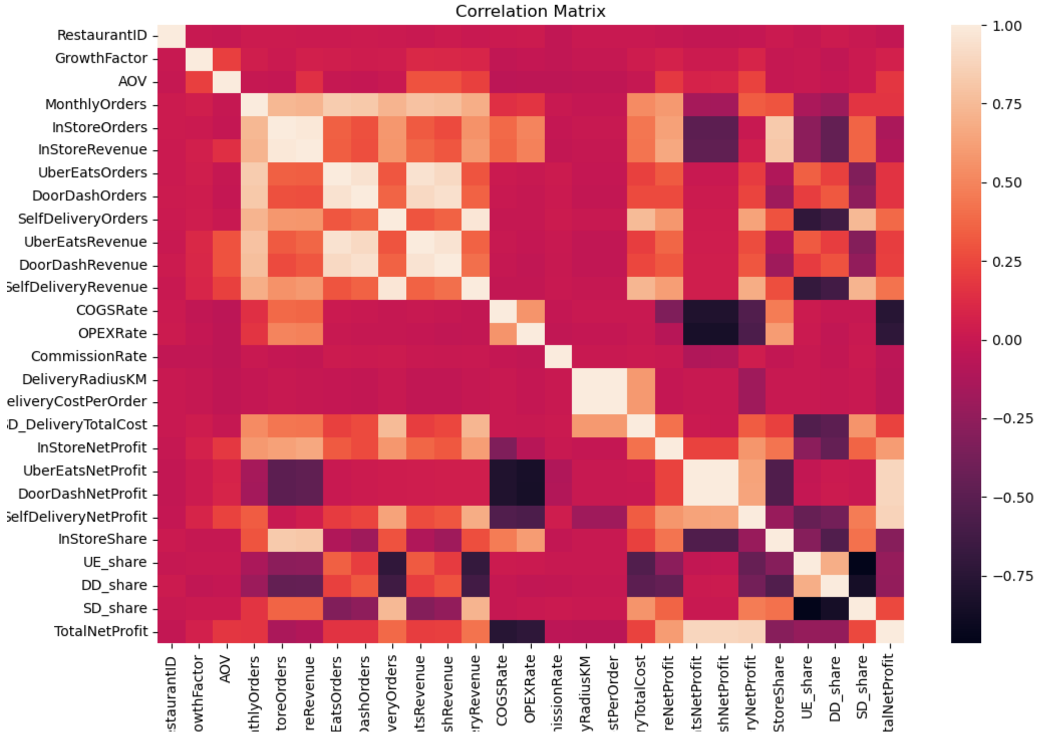
Description:
The chart visualizes the distribution of restaurant orders across InStore, Uber Eats, DoorDash, and Self Delivery channels.

Figure B.2: Profit Distribution Analysis



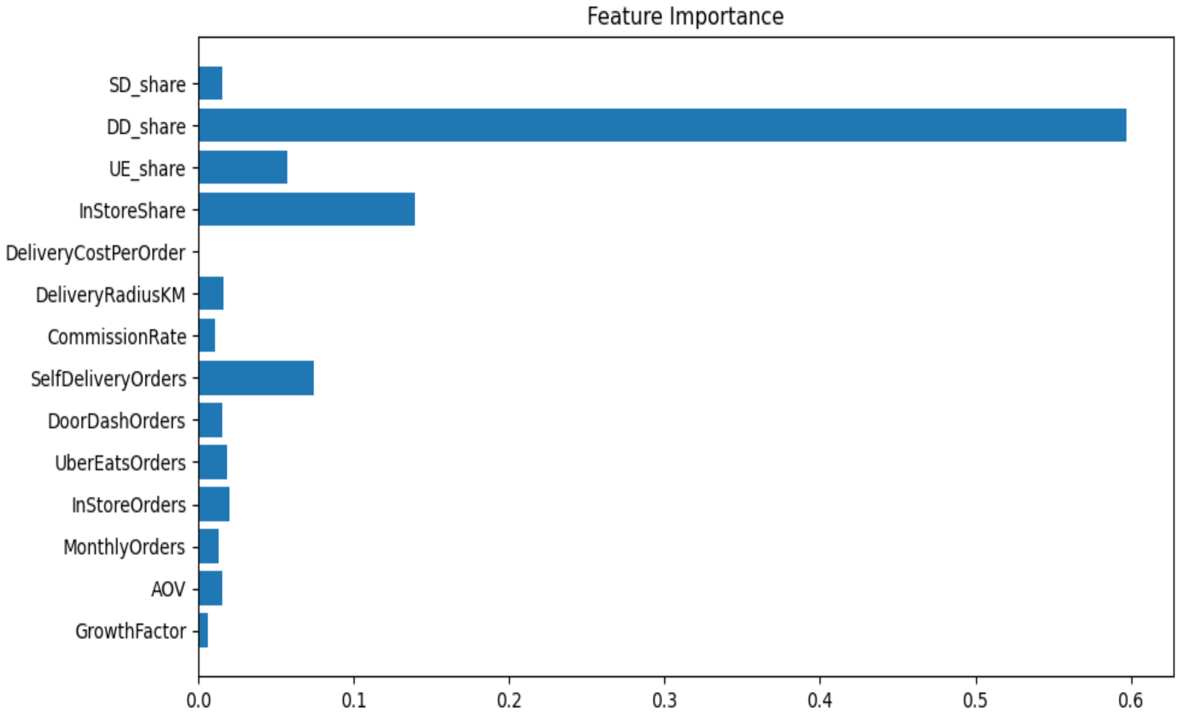
Description:
This graph illustrates the distribution of Total Net Profit across restaurants in the dataset and helps identify profitability patterns.

Figure B.3: Correlation Matrix



Description:
The correlation matrix shows relationships between operational variables and profitability indicators.

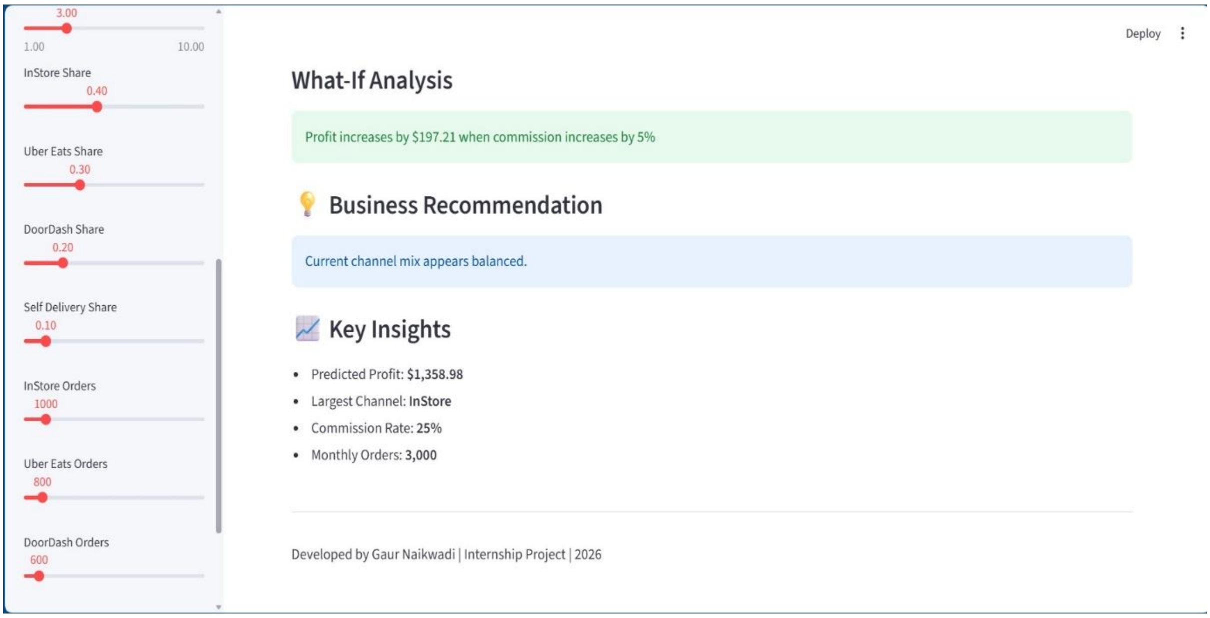
Figure B.4: Feature Importance Analysis



Description:
This figure highlights the most influential variables contributing to restaurant profitability predictions.

Appendix C: Dashboard Features

Figure C.1: What-If Analysis Module



Description:
The What-If Analysis feature allows users to evaluate the impact of operational changes such as commission rate adjustments on predicted profitability.

Figure C.2: Business Recommendation Engine

Description:
This module generates business recommendations based on operational conditions and channel performance.

Figure C.3: Download CSV Functionality

Description:
This feature enables users to export dashboard input configurations for future analysis and reporting.

Appendix D: Project Files

Source Files

1. app.py
2. train_model.py
3. eda.py
4. README.md
5. requirements.txt

Model File

1. profit_model.pkl

Dataset

1. SkyCity Auckland Restaurants & Bars.csv

Appendix E: Tools and Technologies

Tool / Technology	Purpose
Python	Programming Language
Pandas	Data Analysis
NumPy	Numerical Computing
Scikit-Learn	Machine Learning
XGBoost	Predictive Modeling
Matplotlib	Data Visualization
Plotly	Interactive Charts
Streamlit	Dashboard Development
VS Code	Development Environment